Ansible Provisioning of EC2 Nodes

In this assignment, I implemented an Ansible-based automation to configure two AWS EC2 Ubuntu instances as managed worker nodes and install the Nginx web server on both machines. The configuration was performed using an Ansible inventory file and a playbook written in YAML format.

1. Inventory Configuration: I created an inventory file named inventory.yaml to define the target EC2 hosts. The inventory contains a host group called ec2_nodes with two EC2 instances (node1 and node2). For each node, the following connection parameters were specified:

- Public IP address of the EC2 instance
- SSH port 22
- Remote user as Ubuntu
- Private key file path for authentication
- Disabled strict host key checking for seamless automation

This inventory file allows Ansible to securely connect to both EC2 machines using SSH and treat them as managed nodes.

2. Playbook-1 (installs nginx): A playbook named playbook.yaml was written to automate the installation of Nginx. The playbook:

- Targets the host group **ec2_nodes**
- Uses privilege escalation (become: true) to execute tasks as root
- Installs the **nginx** package using the Ansible **apt module**
- Updates the package cache before installation to ensure the latest version is installed

3. Playbook-2 (changes default site): A playbook named **changesite.yaml** was written to automate the modification of the default Nginx webpage. The playbook:

- Targets the host group **ec2_nodes**
- Uses privilege escalation (become: true) to access system directories
- Replaces the default /var/www/html/index.html file with a custom HTML page using the Ansible **copy module**
- Ensures the custom content is deployed automatically on both EC2 worker nodes

```
aaru@Dell:~/ansible-assignment$ ls
changesite.yaml  inventory.yaml  playbook.yaml
aaru@Dell:~/ansible-assignment$
aaru@Dell:~/ansible-assignment$ cat inventory.yaml
ec2_nodes:
  hosts:
    node1:
      ansible_host: 98.81.160.226
      ansible_port: 22
      ansible_user: ubuntu
      ansible_ssh_private_key_file: /home/aaru/elk.pem
      ansible_ssh_common_args: '-o StrictHostKeyChecking=no'
    node2:
      ansible_host: 54.174.97.44
      ansible_port: 22
      ansible_user: ubuntu
      ansible_ssh_common_args: '-o StrictHostKeyChecking=no'
      ansible_ssh_private_key_file: /home/aaru/elk.pem
aaru@Dell:~/ansible-assignment$
```

```
aaru@Dell:~/ansible-assignment$
aaru@Dell:~/ansible-assignment$ cat playbook.yaml
- name: Install and run Nginx on EC2 Ubuntu
  hosts: ec2_nodes
  become: true

  tasks:
    - name: Install nginx
      apt:
        update_cache: yes
        name: nginx
        state: present
aaru@Dell:~/ansible-assignment$
aaru@Dell:~/ansible-assignment$
```

```
aaru@Dell:~/ansible-assignment$ cat changesite.yaml
- name: Change default Nginx site
  hosts: ec2_nodes
  become: true

  tasks:
    - name: Create custom index page
      copy:
        dest: /var/www/html/index.html
        content: |
          <h1>Welcome to My Ansible Managed EC2 Server</h1>
          <p>Configured using Ansible Playbook</p>
aaru@Dell:~/ansible-assignment$
```

Fig: Ansible inventory file and two playbook written in YAML format

4. Execution: The automation was executed using the command:

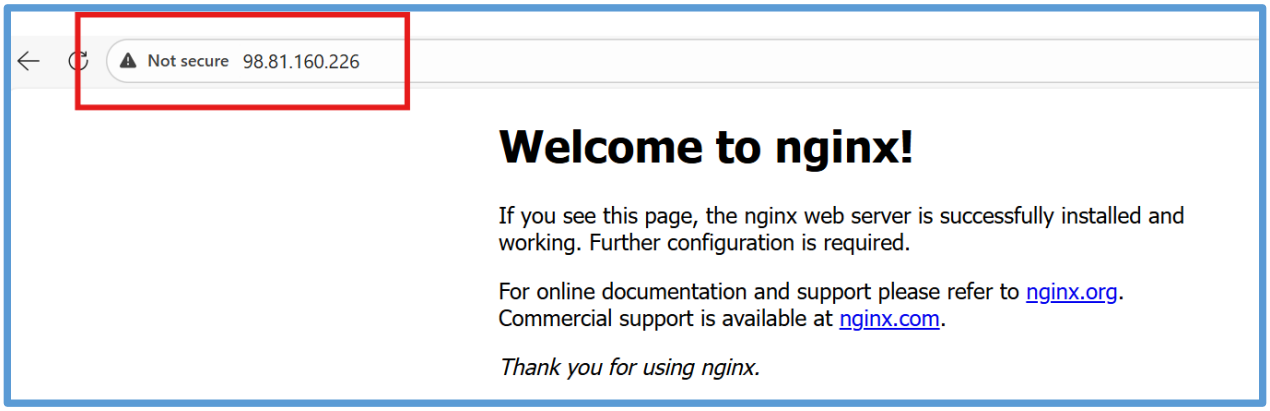
ansible-playbook -i inventory.yaml playbook.yaml

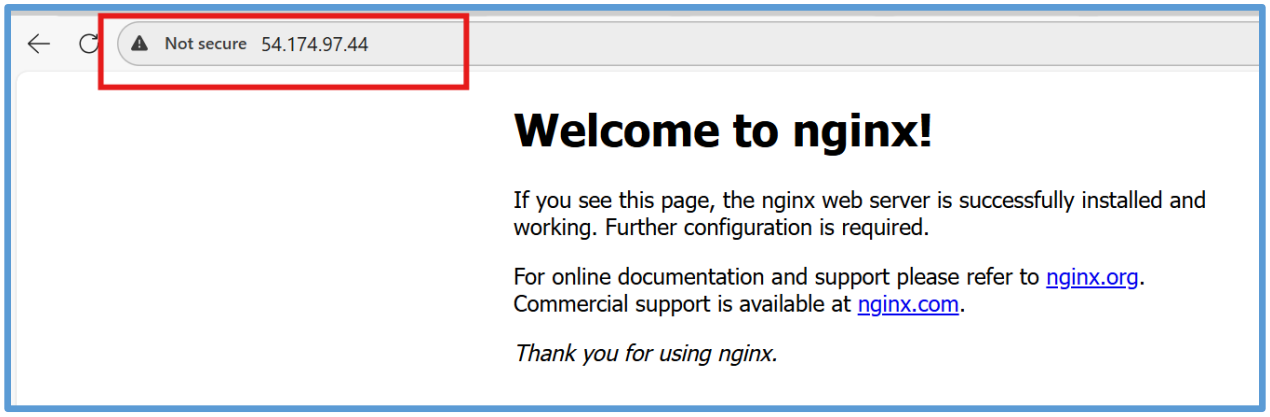
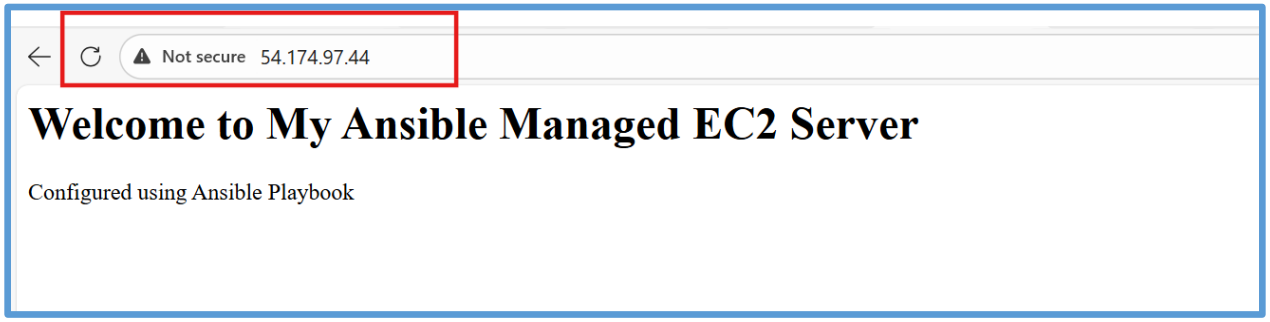
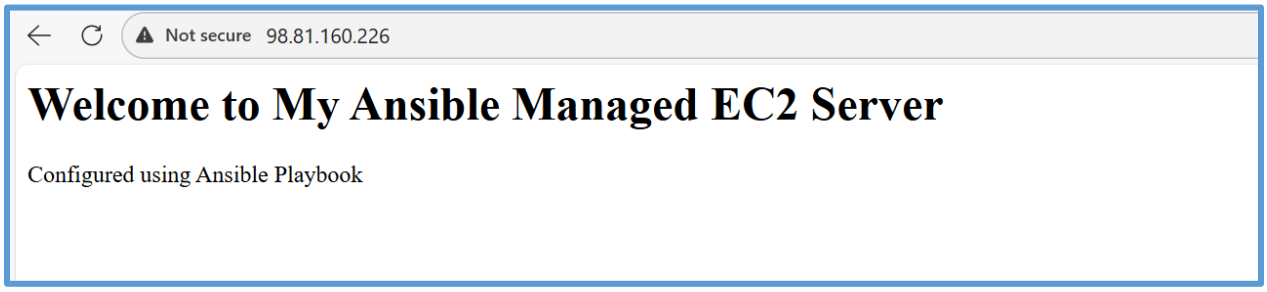
```
aaru@Dell:~$ chmod 400 elk.pem
aaru@Dell:~$ cd ansible-assignment/
aaru@Dell:~/ansible-assignment$ ansible-playbook -i inventory.yaml playbook.yaml

PLAY [Install and run Nginx on EC2 Ubuntu] *************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [node1]
ok: [node2]

TASK [Install nginx] ***********************************************************************************
changed: [node2]
changed: [node1]

PLAY RECAP *********************************************************************************************
node1                      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored
=0
node2                      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored
=0
```

ansible-playbook -i inventory.yaml changesite.yaml

```
aaru@Dell:~/ansible-assignment$ ansible-playbook -i inventory.yaml changesite.yaml

PLAY [Change default Nginx site] ***********************************************************

TASK [Gathering Facts] *********************************************************************
ok: [node2]
ok: [node1]

TASK [Create custom index page] ************************************************************
changed: [node2]
changed: [node1]

PLAY RECAP *********************************************************************************
node1                      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    r
escued=0    ignored=0
node2                      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    r
escued=0    ignored=0
```

POC for nginx:





POC of changing default site: